Tabletop Beat

Your real desk is the drum kit. Your fingers hit real wood and the music answers.

Gaming track | New Experience | Target: Best New Gaming Experience

AT A GLANCE

Track / division	Gaming / New Experience
Primary prize	Best New Gaming Experience ($100,000; runner-up $50,000)
Fallback special award	Boldest Original Concept ($25,000)
Session length	3-minute tracks; a full sitting is 5 to 10 minutes
Room requirement	One horizontal surface: a desk, table, lap tray or airplane tray table
Core tech	WebXR hand tracking (25 joints per hand), plane detection, Web Audio clock, optional depth occlusion
Biggest risk	Detecting finger-to-desk contact precisely enough for rhythm timing
Decision gate	Week-1 parallel spikes: taps (days 1-3), Shadowplay shadows (day 4), Desk Tide depth (day 5)
Availability assumed	Full time: roughly 300 to 400 hours through 17 Nov

The one-line pitch

A mixed-reality rhythm game played by tapping, sliding and slapping your actual desk. Notes glide across the real tabletop toward your hands, and every hit lands on a surface that pushes back. The desk gives hand tracking the one thing it has always lacked: touch.

SECTION 01

The idea

Tabletop Beat turns the surface in front of you into a percussion instrument. When you sit down at your desk and enter the experience, the app finds the tabletop and lays a strip of glowing pads along its near edge, within easy reach of your fingers. Notes appear at the far side of the desk, near your monitor, and slide across the real wood toward the pads. You hit the pads by physically tapping the desk. Fast taps, long holds, finger slides and flat-palm slaps each map to a different note type.

Everything happens on a real surface at real scale. Your fingers stop when they hit the desk, the desk makes its own sound, and the game layers its music on top of that physical impact. The result feels much closer to drumming on a table in time with a song than to poking at a floating interface.

Why this matters to the judges

The competition asks for hands-first experiences, and the most common complaint about hand tracking is that tapping air feels hollow: there is no resistance and no confirmation. Tabletop Beat is built on the observation that **real surfaces supply the haptics that hands lack**. That is a platform-level insight, not only a game theme, and it is exactly the kind of idea the Boldest Original Concept award describes: a core loop that would fall apart on any other platform.

SECTION 02

The core insight: passive haptics

In VR research, passive haptics means using real physical objects to give touch feedback to virtual interactions. Most hand-tracked apps skip it because they cannot assume any particular object exists. Almost every Quest user who sits down has a desk or table in front of them, though, and Quest's scene understanding already knows where it is. That makes the desk the one piece of passive haptic hardware every player owns.

What the real desk provides	Why it improves a rhythm game
Physical resistance	Your finger stops at a consistent height, so taps become repeatable and muscle memory can form.
An acoustic click	The tap makes sound instantly, with zero latency. This masks the small delay before the game's own sound plays.
A stable reference plane	Pad positions never drift. You learn where the pads are by feel, the way a drummer knows a kit.
Rest for your arms	Wrists and forearms can rest on the desk, which avoids the 'gorilla arm' fatigue of mid-air play.
A familiar gesture	Drumming fingers on a table is something everyone already does. No tutorial is needed to explain it.

SECTION 03

What a session looks like

First launch: the first five minutes

Time	What the player sees and does	What the system is doing
0:00	Passthrough fades in. A soft light sweeps across the real desk and settles as a glowing strip along its near edge.	Plane detection finds the surface labeled 'table' closest to the player's hands. If none is found, a hand ray lets the player tap any flat surface instead.
0:20	A pulse of light appears on the strip. The player taps it, and a ripple of light spreads across the real desk with a drum sound.	The first taps teach the core gesture without any text.
0:45	Pulses arrive on a steady beat. The player taps along for 8 beats.	Silent calibration: measures the height of the desk under each fingertip (correcting the detected plane to within a few millimetres) and the player's personal timing offset.
1:00	The first track, 'Warm Up', starts with a single lane.	Chart difficulty is chosen from calibration accuracy.
2:30	Mid-song, a second lane lights up with a flourish.	Progressive disclosure: new mechanics appear inside music, never in menus.
3:30	The song ends. Light gathers into a first star, which drops onto a small shelf at the side of the desk.	Progress saved to browser storage (IndexedDB).
4:00	Three glowing record discs appear on the desk: the next tracks. The player pinches one to play.	The main menu is physical objects on the desk, not a panel.

A normal return visit

1. Put on the headset at the desk. The app remembers the desk position using a saved spatial anchor (a fixed reference point Quest keeps in the real room), so the pads reappear instantly.
2. Play the Daily Groove: a single 90-second challenge generated from today's date, the same for every player.
3. Pick one or two tracks to improve a score or earn a star.
4. Leave at any point. Removing the headset pauses the song, and putting it back on resumes with a 3-beat count-in.

SECTION 04

Mechanics in detail

Note types and how they are detected

Gesture	Musical role	Detection rule (per fingertip, per frame)
Tap	Standard hit (kick, snare, hat, tom)	Fingertip moves downward faster than a threshold and crosses into a band about 1.2 cm above the calibrated desk plane.
Hold	Sustained note (pad, bass)	Tap, then the fingertip stays within the contact band for the note's length.
Slide	Glissando or riser between two pads	Fingertip stays in contact while travelling sideways across pad boundaries, tracked as a path.
Palm slap	Accent or crash	At least three fingertips and the palm centre enter the contact band within 40 ms, **and** the palm is moving downward faster than a threshold. Proximity alone never counts, so a resting hand cannot trigger it.
Two-hand roll	Fill at the end of a phrase	Alternating taps from left and right hands, judged as a group rather than note by note.

The note highway lies on the desk

Most rhythm games put notes on a vertical wall or have them fly at the player. Tabletop Beat lays the highway flat across the tabletop: notes are thin glowing discs that slide from the far edge of the desk to the pads at the near edge. This uses the real depth of the desk (usually 50 to 80 cm) as the timing runway. It keeps all important content in the lower-central part of the view, where the eyes naturally rest when looking at hands, and it stays readable on both the Quest 3 and the narrower Quest 3S field of view.

Lane count scales with difficulty: 1 lane in the tutorial, 3 on Easy, 4 on Normal, and 5 on Hard (4 pads plus a palm-slap zone). Pads sit on an arc that follows the natural reach of the fingers with the wrists resting on the desk.

Timing and judgement

- **The audio clock is the master clock.** Note positions are computed from the Web Audio API's currentTime, never from frame counts, so frame drops never desynchronise the chart.
- **Judgement windows (initial values, tuned in playtesting):** Perfect within 60 ms, Great within 100 ms, Good within 150 ms. These are wider than in controller rhythm games to absorb hand-tracking latency.
- **Contact prediction:** the system uses fingertip speed and distance to the plane to estimate the exact moment of contact, recovering some of the tracking delay.
- **Personal offset:** calibration measures each player's average early/late bias and shifts the windows to compensate. It is saved per device.
- **Output latency:** where the browser reports it, AudioContext.outputLatency is added to the offset. Support in Quest Browser is unverified; if it is missing or reports 0, the calibrated personal offset covers the whole delay.

- **Layered sound:** each hit triggers a sample on top of the backing track. The real tap's acoustic click lands first, which makes the digital sound feel immediate even when it arrives a few tens of milliseconds later.

Modes

Mode	Description	Priority
Tracks	Original songs with hand-authored charts at three difficulties.	Must ship
Daily Groove	A 90-second chart generated from the date. The same for everyone, with a streak counter.	Must ship
Jam	No fail, no chart. The desk becomes a sampler: every pad plays a sound and loops can be layered.	Must ship
Record	Tap along to a song to author your own chart. Saved locally and shareable as a code.	Stretch; supports the reason-to-come-back story and hints at a user-generated-content roadmap

SECTION 05

Why passthrough and the real room matter

The judges state that passthrough should be purposeful: the real room should meaningfully change the experience. In Tabletop Beat the real room is the controller.

- **The desk is the input device.** Without a physical surface the game has nothing to hit. Turning passthrough off would leave the player tapping a desk they cannot see.
- **The playfield adapts to the desk.** The highway length and pad spacing come from the detected desk size. A small laptop table gets a compact 3-lane layout; a big desk gets the full 5 lanes.
- **Your real hands occlude the notes (must ship).** On Quest 3, depth sensing lets real fingers hide the virtual discs passing under them. This sells the illusion that the notes are on the desk and carries the video hook. Some judges may test on a Quest 3S, so the fallback (a soft glow drawn at each fingertip) must look good too.
- **Light spills onto the real desk.** Hit ripples are drawn as additive light on the desk plane, so it looks as if the desk itself is glowing.
- **Clutter-aware setup.** If the depth data shows objects in the lane area (a mug or a phone), the setup screen outlines them and suggests sliding them aside, or shifts the lanes to a clear strip.

Holding up in rooms the developer never tested

The technical criterion explicitly rewards experiences that work in unfamiliar rooms. Tabletop Beat needs exactly one thing: a horizontal surface in front of the player. If Space Setup did not label a table, the player points a hand ray at any flat surface and pinches, and the app places the pads there using a hit test. It works on a kitchen table, a lap desk, a book on the knees and an airplane tray table, which makes the airplane seat test literal.

SECTION 06

Seated, FoV-aware design

- All interaction happens within about 60 cm of the chest, below eye level, at a downward gaze of roughly 25 to 45 degrees.
- The playfield is at most about 45 cm wide at arm's length, which stays inside the narrower Quest 3S field of view without head turning.
- The score and combo float just above the far end of the lanes, in the same view as the notes, never at the edges.
- No standing, no reaching, no turning. Forearms can rest on the desk throughout.
- **Resting hands are safe.** Calibration records each player's resting hand height separately from their tap height, so the heel of a resting hand is never mistaken for a hit.

SECTION 07

Progression and reasons to come back

System	What it does	Why it pulls the player back
Stars and track unlocks	Each track awards up to 3 stars. Stars unlock new tracks and difficulties.	A clear goal for the next session.
Daily Groove streak	A new 90-second challenge every day, identical for all players.	A reason to open the app daily, and a short satisfying moment.
The worn drum skin	The pad surface slowly shows a heat map of where you have hit over days and weeks, like a real drum head wearing in.	Visible physical growth between sessions, unique to this game.
Ghost of your best	A faint ghost hand replays your personal best, so you race your earlier self.	Improvement is shown, not just scored.
Mastery rings	Each track's record disc gains a ring for each difficulty cleared.	Collection instinct; the desk shelf fills up.

SECTION 08

Accessibility

- **One-hand mode:** charts are remapped so every note can be played with one hand.
- **Large-pad mode:** fewer, wider pads for players with limited fine motor control.
- **Tempo control:** play any track at 70 to 100% speed while learning.
- **No-fail option:** the song never stops early.
- **Soft-touch mode:** accepts gentle taps, for quiet environments or shared rooms where loud desk drumming would disturb others.
- **Not colour-dependent:** each lane has its own shape (circle, square, triangle, diamond), so colour-blind players can read the highway.
- **Deaf and hard-of-hearing players:** a visual metronome pulse runs along the lane edges, and all hits give strong visual feedback, so the game is playable with sound off.

SECTION 09

Art direction and sound

Look

Neon-on-wood. The real desk stays visible and uncovered. Virtual elements are thin, bright and emissive: flat discs for notes, crisp outlines for pads, and ripples of light for hits. There are no heavy 3D models to produce, which suits a solo schedule and keeps the frame rate high. Each track has its own two-colour palette so tracks feel distinct without new art.

Sound

- Short, punchy one-shot samples so hits feel immediate.
- Spatial audio positions each pad's sound at its real location on the desk.
- **Eight tracks** of 2 to 3 minutes, commissioned or properly licensed. Self-composed music is only competitive if you already produce music.
- **Licensing for a web app is stricter.** The audio files are publicly downloadable from the hosted site, so the licence must allow distributing them, not just playing them, and must allow competition and promotional use. Keep every licence in the repo.
- Commissioning is the safest route: a work-for-hire agreement that grants the needed rights, with the musician credited in the submission.

SECTION 10

Technical architecture (IWSDK)

Platform features

Feature	Used for	Notes
Hand tracking (WebXR Hand Input)	All input: 25 joint poses per hand every frame	Required. The app never asks for controllers.
Plane detection	Finding the desk (semantic label 'table')	Planes come from the user's Space Setup.
Hit test	Fallback placement on any surface	Pinch on a surface to place pads there.
Anchors	Remembering the desk position between sessions	Persistence across sessions needs checking in Quest Browser.
Depth sensing	Real hands occluding notes; clutter detection	Quest 3 only; optional enhancement.
Web Audio API	Music playback, sample triggering, master clock	Unlock the audio context on the first pinch.

ECS design

IWSDK is built on an ECS (entity-component system): components are plain data attached to entities, and systems run every frame over the entities that match a query. The planned components and systems are:

Component (data)	Fields
DeskSurface	plane pose, width, depth, calibrated height offset
Lane / Pad	index, shape, local position on desk, width
Note	hit time (ms), lane, type (tap/hold/slide/slap), duration, state
FingerContact	hand, finger, distance to plane, velocity, contact state
Judgement	note id, offset ms, grade

System (behaviour)	Responsibility
DeskSetupSystem	Chooses the desk plane or runs the tap-to-place fallback, saves the anchor.
CalibrationSystem	Runs the 8-beat tap-along; corrects plane height and timing offset.
ContactDetectionSystem	Every frame, computes fingertip-to-plane distance and velocity; emits contact start/end events with hysteresis.
ChartPlaybackSystem	Positions notes from the audio clock; spawns from a pooled set of note meshes.
JudgeSystem	Matches contact events to notes; assigns grades; updates combo and score.
FeedbackSystem	Ripples, pad flashes and samples from pre-allocated pools.

System (behaviour)	Responsibility
ProgressionSystem	Stars, streaks, the worn-skin heat map; persists to IndexedDB.

Chart format

Charts are small JSON files: { bpm, offsetMs, lanes, notes: [{ t, lane, type, dur }] }. They are authored by tapping along in a simple desktop tool, or in-headset in Record mode, and then hand-edited.

Performance budget

- Target 72 fps (13.9 ms per frame). The judges' floor is 60 fps.
- At most 64 active notes, drawn with a single instanced mesh.
- All note, ripple and sound objects come from pools created at start-up. Nothing is allocated per frame.
- Contact detection is cheap arithmetic on 10 fingertip positions per frame.

SECTION 11

Risks and mitigations

Risk	Severity	Mitigation
Fingertip height jitter makes contacts unreliable	High	Hysteresis (enter at 1.2 cm, exit at 2.5 cm), a velocity requirement, per-user plane calibration. This is tested first in the week-1 spike.
Hand-tracking latency makes timing feel mushy	High	Wider windows, contact prediction, personal offset, and the real tap sound masking the delay. Charts favour moderate tempos.
Fast motion causes tracking loss	Medium	Design around fingertip taps rather than drum rolls at speed; avoid tempos above about 140 BPM for dense sections.
Hands overlapping confuse tracking	Medium	Lane layout keeps left and right hands on separate halves of the desk.
No table found in the judge's room	Medium	Tap-to-place fallback on any surface, tested from the first week.
Desk clutter in the lanes	Low	Setup outlines obstacles (Quest 3) and lets the player shift the lane strip.
Music licensing	Medium	Commission or license with distribution rights (the files are downloadable from the web app); keep licences in the repo.
Resting wrists trigger phantom palm slaps	High	Slaps require downward speed, not just proximity; calibrate each player's resting height separately. Tested in the week-1 spike with wrists resting and lifted.
Passthrough cannot be captured for the video	High	Verify on day 1 that Quest recording or casting captures passthrough together with the WebXR layer. If not, plan an alternative capture method before week 4.
Glass, glossy or dark desks	Medium	These surfaces can confuse plane detection and depth sensing. Include them in the 10-room playtest; the tap-to-place fallback and plane calibration cover most cases.

Risk	Severity	Mitigation
Tracking quality varies with lighting and skin tone	Medium	Playtest across varied lighting, hand sizes and skin tones; tune thresholds on the whole set, not the developer's hands.
AudioContext.outputLatency missing on Quest Browser	Low	Fall back to the calibrated personal offset.
Persistent anchors unsupported	Low	Re-detect the desk on each launch; it takes about 2 seconds.

Week 1: parallel spikes

With full-time hours, week 1 tests the three ideas that matter in parallel, so the choice is made on evidence rather than preference. More hours reduce scope and content risk, but not platform risk: tracking latency, depth resolution and thumb occlusion do not improve with effort. That is what these spikes measure.

Day	Spike	What to measure
1	Passthrough capture check (first hour)	Record a short clip of a WebXR scene over passthrough using Quest recording or casting. If it fails, the video plan changes.
1 - 3	Tabletop Beat taps	End-to-end latency: film the desk through the headset lens with a 240 fps slow-motion camera and count frames from physical contact to the virtual ripple. Detection rate with wrists resting and with wrists lifted. Timing spread (standard deviation) at 90, 120 and 140 BPM.
4	Shadowplay shadow render	Project simple joint capsules from a lamp onto the detected wall and judge on device whether the shadows look convincing.
5	Desk Tide depth probe	Can a mug and a book be resolved in a height map at desk distance, on Quest 3 and on Quest 3S?

Go criteria for Tabletop Beat

At least 95% of real taps detected and at most 2% phantom contacts, **both with wrists resting and with wrists lifted**. A timing spread under about 25 ms after calibration at 90 and 120 BPM (140 BPM may be looser; if so, cap dense sections below it). And it has to feel satisfying to play.

Decision order after week 1

1. **Taps pass:** build Tabletop Beat at full scope.
2. **Detection passes but timing precision fails:** loosen timing with groove-style scoring and wider windows. This costs about 2 days; still build Tabletop Beat.
3. **Detection itself fails:** switch to Shadowplay, which now has enough hours for its content load (script, voice and five chapters).
4. **Desk Tide** only if the depth probe is clearly excellent. It has the highest ceiling, but its outcome depends on sensor physics, not effort.

SECTION 12

Seven-week build plan

Week	Dates	Goal	Exit criterion
Week 1	29 Sep - 4 Oct	Parallel spikes: passthrough capture, taps, Shadowplay shadows, Desk Tide depth	Idea chosen against the decision order; capture method confirmed
Week 2	5 - 11 Oct	Desk setup with fallback; chart playback on the audio clock; all note types on 4 lanes	One test song playable with tap, hold, slide and slap judged correctly
Week 3	12 - 18 Oct	Calibration (tap height, resting height, timing offset); first five minutes; stars; saves	A new player reaches a first star with no instructions
Week 4	19 - 25 Oct	Depth occlusion and 3S fallback; Jam mode; soft-touch mode; 4 tracks; **start capturing video footage**	First usable footage recorded; half the tracks in
Week 5	26 Oct - 1 Nov	Daily Groove; worn-skin heat map; ghost replay; tracks 5 to 8; accessibility options	Content-complete for the must-ship list
Week 6	2 - 8 Nov	Playtests in 10+ rooms (varied lighting, skin tones, hand sizes, glass/glossy/dark desks); performance pass; Record mode if ahead	72 fps held; setup and detection work in every tester's room
Week 7	9 - 15 Nov	First-five-minutes iteration until every cold tester succeeds; sound mix; final video edit; submission text	Video cut from weeks of footage; build frozen candidate
Ship	16 - 18 Nov	Final checks on a clean device; submit on 17 Nov	Submitted a day early; hosted link verified

Where the extra full-time hours go

Spend them where judges score, not on features judges will not see: **playtests** in 10 or more real rooms, **iterating the first five minutes** until a cold tester reaches a first star with zero instructions every time, and **video footage** captured from week 4 onward so the final cut has real choices.

SECTION 13

Scope

Must ship	Nice to have	Cut
Desk detection with fallback; tap/hold/slide/slap; 8 commissioned or licensed tracks; calibration including resting height; stars; Daily Groove; depth occlusion (Quest 3) with a polished 3S fallback; worn-skin heat map; ghost replay; Jam mode; soft-touch, one-hand and tempo options	Record mode with shareable chart codes (stretch)	Online leaderboards, multiplayer and accounts: a backend adds failure modes during the judging window and adds nothing to the score. Song import from files.

SECTION 14

Demo video plan (under 3 minutes)

Time	Beat	Shot
0:00 - 0:10	Hook	Close-up of real fingers drumming a real desk in passthrough, light exploding from each impact. No title card first.
0:10 - 0:30	The insight	Side by side: tapping air (nothing happens, finger wobbles) versus tapping the desk (crisp stop, ripple). Caption: 'Your desk is the haptics.'
0:30 - 1:10	First five minutes	Real-time footage of setup, calibration taps and the first song, showing no text walls.
1:10 - 1:50	Depth of play	Hard chart with slides and slaps; combo climbing; the worn drum skin after a week.
1:50 - 2:20	Anywhere	Montage: kitchen table, lap desk, a tray table, all in real rooms.
2:20 - 2:50	Close	Daily Groove streak, the star shelf filling, final logo on the desk.

SECTION 15

Submission draft

Name	Tabletop Beat
Tagline (under 140 characters)	A mixed-reality rhythm game where your real desk is the drum kit. Tap, slide and slap real wood; the music answers.
Track / division	Gaming / New Experience
Target launch	Public web release Q1 2027; Horizon Store (PWA) Q2 2027

Description (first draft)

Hand tracking has one big weakness: tapping air feels like nothing. Tabletop Beat fixes that by using the one piece of haptic hardware every Quest owner already has, the desk in front of them.

When you sit down, Tabletop Beat finds your real tabletop using Quest's scene understanding and lays glowing pads along its edge. Notes slide across the real wood toward your fingers. You tap, hold, slide and slap the actual desk to play. Your fingers stop on a real surface, the desk makes a real sound, and light ripples across it with every hit.

We built it with the Immersive Web SDK. Contact detection measures each fingertip's distance to a calibrated desk plane and uses velocity and hysteresis to detect hits; the Web Audio clock drives every note so timing never drifts. The playfield adapts to the size of your desk, and if no table is found you can pinch any flat surface to play there, including an airplane tray table.

There are no controllers and no menus to learn: the first minute teaches itself through tapping. A daily challenge, star progression and a drum skin that slowly wears in where you play give you a reason to come back tomorrow. Next: a chart editor for the community, two-player desk jams, and a store release.

SECTION 16

How it maps to the judging criteria

Criterion (25% each)	Evidence in Tabletop Beat
Innovation and Creativity	Passive haptics from the user's own desk; a genre (rhythm) re-imagined around real surfaces rather than mid-air gestures.
Experience Design	Seated by nature, teaches itself in under a minute, passthrough is essential, all content in the lower-central view.
Technical Implementation	Plane detection with a robust fallback, custom contact detection, audio-clock timing, optional depth occlusion, pooled rendering at 72 fps.
Polish and Presentation	Minimal emissive art that is achievable solo and looks sharp on video; a hook that reads in the first 5 seconds.

Hard requirement and design guidelines

Test	What the rules ask	How this idea passes
Hands-first (hard requirement)	Complete the entire experience without ever pairing a controller.	Every action is a finger or palm touching the desk; menus are physical record discs picked up with a pinch.
Airplane seat test	Every interaction works within a two-foot radius, seated.	The entire playfield sits on the desk within forearm reach; arms can rest throughout.
One bus stop test	Fast cold start, clean pause/resume, a satisfying moment in 10 minutes or less.	Cold start is about 10 seconds with a saved anchor; one track is 3 minutes; taking the headset off pauses.
Take it away test	Remove any third-party service and a project still stands.	No third-party service at all. Everything runs locally in the browser.

Competition rules that shape this plan

- **One prize per entry.** Rule 7 limits each entry to a single prize, so this brief aims at one track prize and names one special award as a fallback rather than trying to stack awards.
- **Submission:** a public HTTPS link (GitHub Pages, Vercel or similar) that stays live until the winner announcement (about 11 Dec 2026), a public YouTube/Vimeo video under 3 minutes, and the Devpost form (name, 140-character tagline, track, division, description of about 500 words, hand-interaction notes, target launch date).
- **No changes after the deadline.** Judging uses the build as it stood at 18 Nov 2026, 12:00 PT. Plan to submit on 17 Nov and freeze the hosted build.
- **Content rules:** English (or English subtitles), no identifiable real people other than entrants, no brand names or logos, all music, art and fonts properly licensed or original.
- **Video rules:** real footage from a Quest or the XR simulator; do not lean on AI-generated video to carry the pitch.
- **Performance floor:** judges expect at least 60 fps on Quest hardware. Target 72 fps.
- **Music:** keep written proof of licence for every track and sample; the sponsor may ask for it.

SECTION 17

Competitive landscape

Existing product type	How Tabletop Beat differs
Controller rhythm games (slash or punch to the beat)	Played with fingers on a real surface, seated, controller-free, with real touch feedback.
VR drum simulators	No virtual kit to hit in mid-air; the player's own desk is the instrument.
MR apps that overlay a real instrument (for example piano tutors)	Needs no instrument at all. Any table works, which is why it can reach every Quest owner.

SECTION 18

Honest weaknesses

- The whole idea depends on contact detection feeling good. If the spike fails, the concept fails; that is why week 1 is spent measuring it before anything else is built.
- Rhythm games need music, and music is the hardest asset for a solo developer to make well. Commission or license it early, with distribution rights.
- Hand tracking caps the speed and density of charts, so expert players may find the top difficulty less demanding than in controller games.
- Desk drumming is loud for people nearby. Soft-touch mode (now in scope) reduces this but changes the feel.
- Resting wrists and palm-slap detection pull against each other; if calibration cannot separate them, palm slaps may need a distinct gesture.

SECTION 19

Scorecard and verdict

Dimension	Score / 10	Comment
Originality	9	A genuine platform insight, not a reskin.
Only possible in MR	10	Without the real desk there is no game.
Works in untested rooms	9	Needs one flat surface; has a fallback.
Solo feasibility in 7 weeks	8	Full-time hours cover content and polish; contact detection is the remaining hard part.
Prize ceiling	$100k	Top Gaming prize; Boldest Original Concept as a fallback.

Verdict

The top recommendation of the six ideas, and full-time availability does not change that: hours were never its bottleneck. Run the week-1 spikes. If the taps pass, commit to the full scope above: the idea is strong, the art is achievable, and the extra time goes into content, playtesting and the video.